

Soham Nighojkar

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EDUCATION

Northeastern University

Expected: May 2028

Candidate for Bachelor of Science in Computer Science

GPA: 3.8/4.0

Honors: Khoury College Dean's List

Relevant Coursework: Object-Oriented Design, Algorithms & Data Structures, Discrete Structures, Foundations of Cybersecurity, Foundations of Data Science, Logic and Computation, Computer Systems, Theory of Computation

Activities: TAMID Group Tech Consulting, NU Impact, Northeastern Club Basketball

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C#, C++

Frameworks: Spring Boot, .NET, Node.js, FastAPI

Libraries: PyTorch, Pandas, NumPy, Apache Spark

Tools & Platforms: AWS, Docker, Kubernetes, Kafka, Apache Airflow, NATS, Prometheus, Grafana, Git, CI/CD, REST APIs, Unit Testing, End-to-End Testing

EXPERIENCE

Philips

Cambridge, MA

Software Engineering Intern

January 2026 – June 2026

- Achieved 100% audit trace coverage across 10+ patient monitoring microservices by redesigning NATS JetStream's advisory subscription model with message subject filtering to catch previously missing logs.
- Eliminated manual metric registration across 20+ microservices by building a self-registering Prometheus system in C# (thread-safe singleton, idempotent deduplication), letting 75+ NuGet packages register metrics independently.
- Cut deployment time 16% and removed manual version bumping by consolidating 18+ fragmented Git workflows across 15 repos into one CI/CD pipeline with release-please versioning and automated Docker tagging.
- Enabled real-time detection of changes to patient health data across 1,000+ connected devices by deploying and monitoring Kubernetes clusters (Helm) on Philips HealthSuite's NATS messaging infrastructure.

Observe.ai

Redwood Shores, CA

Software Engineering Intern

June 2025 – August 2025

- Reduced client-side timeouts 58% by building a token-bucket rate limiter with a local cache and a Redis (Redisson) fallback, sustaining 25,000+ daily REST/WebSocket requests.
- Maintained accurate state across 1,000+ concurrent WebSocket connections while reducing stale connections 22%, via a connection-limiting layer using a handshake interceptor and atomic Redis counter.
- Delivered sub-3-second transcription latency across 5,000+ calls by engineering a Kafka pipeline that streamed WebSocket audio to Deepgram STT and fanned transcripts out to multiple downstream consumers.

PROJECTS

Signal Trading Engine | *Python, FastAPI, TypeScript, PostgreSQL, Redis, Docker, Pandas, Statsmodels*

- Built a full-stack quant research platform (FastAPI + Redis/RQ async workers) running configurable multi-ticker backtests over 20-year lookbacks, offloading heavy computation off the request path to keep API latency low.
- Powered the platform's signals with a statistical engine computing pairwise cointegration and correlation across up to 200 ticker pairs, gated by a Dockerized GitHub Actions pipeline running 66+ tests per merge to catch signal logic regressions.
- Developed a TypeScript/Next.js frontend rendering ranked signal tables and per-pair equity curves that overlay train and out-of-sample performance (Recharts), validated client-side against the backend's parameter bounds.

Crypto Anomaly Detector | *Python, Apache Kafka, Apache Airflow, FastAPI, PySpark, PostgreSQL, Redis, Docker*

- Flagged suspicious trades in under 100ms for 99% of requests by designing a real-time market manipulation pipeline (Kafka, PySpark Structured Streaming, FastAPI) that sustained 300+ messages/sec.
- Automated model retraining end-to-end with an Apache Airflow nightly DAG that promotes only non-regressing models (within 0.01 F1), plus a weekly drift monitor flagging feature distribution shifts within 7 days.
- Designed a 3-layer PyTorch MLP (128 and 64 hidden units, class-weighted BCE) scoring four features, including a small-trade ratio that flags wash trading, with the alert threshold tuned for max F1 under a 5% false-positive cap.